THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two numbers are in the ratio 4 : 7 and their sum is 99. The larger number is: 1

(a) 72 **(b)** 63 **(c)** 49 **(d)** 36

2. Two numbers are in the ratio 3 : 5. If 4 is added to each, the numbers become 28 and 44. The original smaller number is: 1

(a) 48 **(b)** 20 **(c)** 24 **(d)** 28

3. A mixture of 60 L has milk and water in the ratio 7 : 3. The quantity of water is: 1

(a) 23 L **(b)** 13 L **(c)** 18 L **(d)** 42 L

4. 6 pipes fill a tank in 10 hours. 4 such pipes will fill it in: 1

(a) 12 hours **(b)** 16 hours

(c) 15 hours **(d)** 6.666666666666666666666666667 hours

5. A : B = 2 : 3 and B : C = 6 : 5. If A + B + C = 165, then C equals: 1

(a) 44 **(b)** 66 **(c)** 33 **(d)** 55

6. On a map with scale 1 : 50000, a distance of 4 cm represents: 1

(a) 20 km **(b)** 6 km **(c)** 200 km **(d)** 2 km

7. 36 km/h expressed in m/s is: 1

(a) 0.6 m/s **(b)** 10 m/s **(c)** 12 m/s **(d)** 129.6 m/s

8. Two numbers are in the ratio 5 : 8 and their sum is 78. The larger number is: 1

(a) 48 **(b)** 54 **(c)** 39 **(d)** 30

9. Two numbers are in the ratio 2 : 7. If 3 is added to each, the numbers become 15 and 45. The original smaller number is: 1

(a) 15 **(b)** 9 **(c)** 24 **(d)** 12

10. A mixture of 84 L has milk and water in the ratio 5 : 2. The quantity of water is: 1

(a) 60 L **(b)** 24 L **(c)** 29 L **(d)** 19 L

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Two numbers are in the ratio 5 : 9 and their sum is 98. Find the numbers. 2

12. Convert 45 km/h into m/s. 2

13. A map has scale 1 : 50000. Two places are 6 cm apart on the map. Find the actual distance in km. 2

14. Two numbers are in the ratio 4 : 11 and their sum is 90. Find the numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A mixture of 40 L contains milk and water in the ratio 7 : 1. How much water must be added to make the ratio 7 : 3? 3

16. A : B = 1 : 2 and B : C = 6 : 5. If A + B + C = 150, find the three numbers. 3

17. The ages of a mother and daughter are in the ratio 7 : 3. The mother is 42 years old. Find their ages 4 years ago and the ratio then. 3

OR

A mixture of 60 L contains milk and water in the ratio 3 : 2. How much water must be added to make the ratio 3 : 4? 3

18. A : B = 3 : 4 and B : C = 8 : 5. If A + B + C = 143, find the three numbers. 3

OR

The ages of a mother and daughter are in the ratio 5 : 2. The mother is 40 years old. Find their ages 6 years ago and the ratio then. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. 12 men can complete a piece of work in 15 days. 5
(i) How many days will 18 men take?
(ii) How many men are needed to finish it in 9 days?

OR

Anil and Bhanu start a business with ₹ 15000 and ₹ 25000. They share a profit of ₹ 800 in the ratio of their investments. Find the ratio of investments and each person's share. 5

20. 20 men can complete a piece of work in 24 days. 5
(i) How many days will 16 men take?
(ii) How many men are needed to finish it in 30 days?

OR

Anil and Bhanu start a business with ₹ 12000 and ₹ 18000. They share a profit of ₹ 1500 in the ratio of their investments. Find the ratio of investments and each person's share. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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